Contribution

Camille: For Team Corndoggers, I completed Sections 7, 8, and 9, focusing on data normalization and scaling. In Section 7, I applied logarithmic transformations to three skewed columns to reduce extreme values. In Section 8, I implemented MinMax scaling to bring numerical features into a consistent range of 0 to 1. In Section 9, I used StandardScaler to standardize numerical features for machine learning. I also reviewed the results to ensure the transformations were applied correctly and helped prepare our Netflix dataset for future predictive models.

Rigoberto: My contribution to the group midterm project was completing Sections 5 and 6, which focused on encoding categorical variables and feature engineering. I used ordinal encoding to convert subscription types into numerical values based on their order: Basic, Standard, and Premium. I also used one-hot encoding to convert categorical features such as gender, region, device, payment method, and favorite genre into numerical columns that a machine learning model could process. For feature engineering, I created five new features: hours per profile, fee per profile, an inactivity flag, a low-engagement flag, and age cohorts. These features were designed to help identify patterns in customer behavior that could be related to Netflix customer churn. I also helped remove redundant columns, including the customer ID and monthly fee, to prepare the dataset for future machine learning modeling. My work helped transform the original customer data into a more useful format for analysis and prediction.

Min Ma Kha

I was responsible for obtaining the information in addition to ensuring that it remained reliable at all stages. I obtained a raw file from Kaggle; I recorded its MD5 hash to guarantee that it was not altered after acquiring the file; and I wrote all loading and analysis cells to define what information we are dealing with, how many rows there are, what data types are used, and what type of classification we are going to use.

The split was my major input into the notebook, which is exactly why it is the ninth cell of it. I tried to do it before performing any analysis since the assignment requires EDA on training data only, and it would have leaked the test data if we followed the classical approach.

I found Section 4 the most fascinating part. The results of Chris's EDA showed that there were no missing values, which appeared to be a problem according to the rubric. I did a second audit to check if there were missing parts concealed as placeholders and duplicate rows, but everything was fine here as well. The problem was solved by the box plot that depicted avg_watch_time_per_day of 98.42, which is simply impossible.

I wrote the coding which sets the limit for 24 hours, changes the 8 offending training rows to NaN, inserts the watch_time_was_imputed marker, and fills them with a coefficient of the training median equal to 0.29. I made a choice of median instead of mean and explained also why I had done it since the mean of this column is 0.80, inflated precisely by values we were trying to correct.

Chris: I was responsible for Section 3 (Exploratory Data Analysis) and Section 11 (Machine Learning Problem and Limitations) of the Netflix Customer Churn and Engagement Analytics project. In Section 3 I created summary statistics, value counts for the categorical values, a check for missing values, and for duplicates. The main features were watch hours, days since last log in, age, subscription type and monthly fee, average daily watch time. Potential problems with the data were found, for example, unrealistic average values for daily watch time and possible redundancy between subscription type and monthly fee. Section 11 defines the problem of customer churn as a supervised binary classification problem. Models that could be used for comparison are described, such as Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, and Support Vector Machine. The corresponding evaluation of metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC are also described. Problems and limitations of the project are described, for example the data could be simulated, a time-based analysis is not possible, and there might be some redundancy in the features.